

Comparing acoustic classification methods to estimate krill biomass in the Georges Bank region from 1999 to 2012

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Abstract

Krill are a critical component of northern-latitude ecosystems, yet are poorly monitored in the Gulf of Maine and Georges Bank regions of the northeast United States continental shelf. We applied five acoustic classification methods (multifrequency single-beam index [MFSBI], dB-differencing, multifrequency index [MFI], and two variations of the Z-score method) to acoustic data collected during surveys conducted annually in the Georges Bank region in order to map krill spatial distribution from 1999 to 2012. Material properties (sound speed and mass density) of krill and length data from net samples collected in the Gulf of Maine and Georges Bank regions were used in conjunction with distorted wave Born approximation (DWBA) predicted target strengths to scale acoustic data to estimates of biomass for the Georges Bank region. All methods require a priori information, ranging from training sets (i.e., empirically based classification) to validated acoustic models of target strength (i.e., theoretically based classification) for the species and acoustic frequencies of interest. Incorporating predicted backscatter from theoretical models can improve confidence in classification, especially when clean training sets are not available, but does come at the cost of a validated acoustic backscatter model. The biomass estimates from the four acoustic classification methods were comparable among each other, with the MFSBI method consistently estimating lowest biomass and the empirically based Z-score consistently estimating highest biomass. Comparisons to net-based estimates were inconsistent, most likely due to net inefficiency (low catchability), targeting of layers, and scaling point estimates to regions.

Krill (especially *Meganyctiphanes norvegica* and *Thysanoessa* spp.) are an important component of many northern marine ecosystems (Tarling et al. 2010), providing a critical energy transfer link between lower (Cleary et al. 2012) and upper trophic levels (Simard and Harvey 2010; Smith and Link 2010). In spite of the recognition as an important component of pelagic food webs, targeted krill monitoring in the western North Atlantic has been sporadic and limited to local efforts. A broad-scale net survey of zooplankton in the Gulf of Maine and Georges Bank regions has been conducted by the National Oceanographic and Atmospheric Administration's (NOAA's) Northeast Fisheries Science Center (NEFSC) since the late 1970s (Kane 2007), but the sampling gear (61-cm diameter bongo frame with 0.333-mm mesh nets) used is inefficient (catchability estimates of <

0.2) for *M. norvegica*, which is a dominant species of krill in these regions (Warren et al. 2003; Stevick et al. 2008). There appears to be a clear need for a time series of regional-scale estimates of abundance and biomass of this important resource.

Acoustic technology and methods have the potential to monitor the spatial distribution and estimate biomass of krill at the regional scales of the Gulf of Maine and Georges Bank (e.g., Simard and Lavoie 1999; McQuinn et al. 2013), as has been done in the Antarctic for many years (e.g., Reiss et al. 2008). Echosounders mounted on sea-going vessels provide a continuous record along the vessel's track of the acoustic scattering throughout the water column at the insonified frequencies. Utilizing multiple frequencies (i.e., multiple echosounders) allows automated discrimination and classification of the acoustic backscatter to categories of fish and zooplankton (e.g., Korneliussen et al. 2016), which greatly increases the efficiency at which large volumes of data collected over multiple years can be processed and analyzed. Methods and algorithms based on the acoustic frequency response of fluid-like scatterers, such as krill, have been used for more than two

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decades to classify backscatter to zooplankton (e.g., Madureira et al. 1993; Stanton et al. 1998). A variety of classification methods are presently available for krill, ranging from approaches where very little advance knowledge of the system and community composition is needed beyond the generic scattering frequency response typical of krill (the multifrequency single-beam index [MFSBI] [Jech and Michaels 2006; Wall et al. 2016], “dB-differencing” [Madureira et al. 1993; Kang et al. 2002], and the multifrequency index [MFI] [Trenkel and Berger 2013]) to approaches based on analysis of pre-existing training sets of known krill scattering (e.g., the Z-score [De Robertis et al. 2010; Ressler et al. 2012]).

We exploited acoustic and midwater trawl data collected during surveys of Atlantic herring (*Clupea harengus*) as well as net data collected during experimental cruises in the Georges Bank region to map the spatial distribution and estimate biomass of krill from 1999 to 2012. Five classification methods were applied to the acoustic data, and net data (length distributions) in conjunction with predicted target strength values from the distorted-wave Born approximation were then used to scale acoustic backscatter to biomass. Biomass estimates from the four acoustic classification methods were compared among each other, and acoustically based estimates were compared to net-based estimates.

Methods

Survey area

Acoustic and biological data were collected as part of the NEFSC's annual acoustic/trawl survey for Atlantic herring (*C. harengus*) conducted in September through October in the Gulf of Maine and Georges Bank regions from 1999 through 2012 (Jech and Michaels 2006; Overholtz et al. 2006). This survey was designed to encompass the pre-spawning and spawning aggregations of Atlantic herring and to derive abundance and biomass estimates of the herring stock in the Georges Bank region. Over the years the survey systematically covered the area from the shoals of Georges Bank, the northern flank of Georges Bank, and north to the deep basins of the Gulf of Maine and west through the Great South Channel to the shores of Cape Cod, Massachusetts. Because the surveys had a random starting location and due to occasional logistical issues over the years, a “common” region (CR) was created (Jech and Sullivan 2014) based on the spatial and temporal distributions of pre-spawning Atlantic herring where over 90% of the aggregations were consistently found within 40 nmi to the north of and 10 nmi to the south of the 90-m bathymetric contour (Jech and Stroman 2012) (Fig. 1).

Length data

Krill length data were obtained from multiple sources, including finfish stomach contents sampled during the NEFSC autumn bottom trawl survey (Smith and Link 2010; Link et al. 2012) and the NEFSC Atlantic herring acoustic/midwater

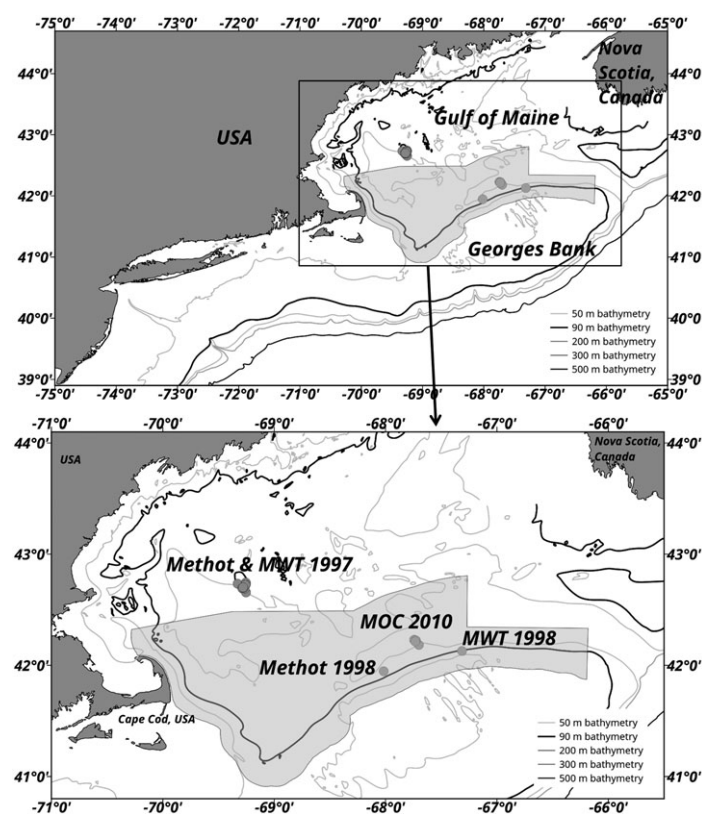


Fig. 1. Net (Methot, midwater trawl [MWT], and multiple opening/closing net and environmental sampling system [MOC]) sampling locations denoted by gray circles; and the “common” survey area encompassing the annual Atlantic herring surveys denoted by the lightly shaded area covering the northern flank of Georges Bank and southern Gulf of Maine.

trawl survey (Jech and Michaels 2006; Overholtz et al. 2006), as well as krill catch data from midwater trawl and Methot net sampling in the Gulf of Maine and Georges Bank regions during the NEFSC Atlantic herring acoustic/midwater trawl survey. The midwater trawl was a high-speed midwater rope trawl (HSMRT) with $\frac{1}{4}$ " knotless cod-end liner designed to sample Atlantic herring and other pelagic fish species (Jech and Michaels 2006; Jech and Sullivan 2014). The Methot net (Methot 1986) was a solid-frame net with a mouth opening of 5 m² and 1 × 2 mm (modified from 2 × 3 mm in Methot [1986]) oval net mesh. The Methot net was towed double-obliquely from the surface to within 4 m of the bottom and back to the surface at a ship speed of 2–2.5 kts.

In addition, krill lengths and length-to-width ratios were measured from depth-stratified MOCNESS (Multiple Opening/Closing Net and Environmental Sampling System [Wiebe et al. 1985]) samples collected in September of 2010 in the Georges Bank region as part of a process study of krill behavior (Fig. 1). The 1-m² MOCNESS was equipped with 333 μ m mesh nets and an active strobe system to mitigate avoidance behavior, and only net samples collected with the strobe on were considered here (Wiebe et al. 2013). Two tows (MOC1 and 2)

were double-oblique from the surface to within 10 m of the bottom, with one net sampling on the down-cast and the other eight nets sampling discrete depth intervals on the up-cast, while an additional two tows (MOC3 and 4) sampled directly within acoustically observed krill layers, with multiple nets collecting replicate samples through the layer.

Krill standard length 1 (SL1 [Mauchline 1980]) was measured and rounded to the nearest millimeter and transformed to the non-standard “acoustic” length (Lawson et al. 2006) for the acoustic models. SL1 was used in the length-weight relationship of McQuinn et al. (2015) for *M. norvegica* in the Gulf of St. Lawrence: W (mg) = $0.007943 \times SL1^{2.947}$. Acoustic length was used for the predictions of acoustic backscatter by the distorted-wave Born approximation (DWBA) acoustic scattering model. The relationship between SL1 and acoustic length (L_A) was: $L_A = 0.91 \times SL1$ based on measurements of a subset of *M. norvegica* sampled by the MOCNESS (Wiebe et al. 2013). Acoustic-length density distributions at 1-mm intervals from 1 mm to 50 mm were generated for each net catch. These density distributions were then combined by gear type and year, then for all density distributions by summing the densities at each 1-mm interval and dividing by the sum of the densities (i.e., normalizing the densities) so that the sum of the normalized densities was one. This combined acoustic-length density distribution ($d_L(L_A)$) was used to weight the DWBA predicted target strengths by acoustic length.

Acoustic model predictions

The DWBA-based acoustic model of krill (Lawson et al. 2006, 2008) was used to predict acoustic backscattering cross-sectional area (σ_{bs} , m²) and target strength (TS, dB re m²) ($TS = 10\log_{10}(\sigma_{bs})$) (Clay and Medwin 1977) as a function of acoustic length (L_A) at frequencies of 12 kHz, 18 kHz, 38 kHz, and 120 kHz ($TS(f, L_A)$). This model represents the shape of the animal as a uniformly bent and slightly tapered cylinder (Stanton et al. 1998; Stanton and Chu 2000; Lawson et al. 2006) and involves multiple parameters. The length of the equivalent cylinder was taken as the “acoustic” length described above (i.e., the distance from the anterior tip of the eye to the end of the sixth abdominal segment [Lawson et al. 2006]). A mean length-to-width ratio for the assumed cylindrical shape of 9.1 was used, measured for a sub-sample of 20 animals sampled by MOCNESS in 2010 (Wiebe et al. 2013). The ratio of the radius of curvature to the length of the cylinder was assumed to be 3 and the taper parameter set to 10 (Lawson et al. 2006), although scattering is mostly insensitive to such details of the shape description when averaging over a distribution of orientations that includes broadside incidence (Stanton and Chu 2000). Acoustic material property measurements (sound speed contrast = 1.0425 and mass density contrast = 1.0343) made in November 2016 followed the methods of Chu and Wiebe (2005) (and references therein) for *M. norvegica* sampled in the western Gulf of Maine (G. Lawson and S. Sakin, unpubl. data). TS was predicted for

acoustic lengths from 1 mm to 60 mm at 1-mm increments, where the length-to-width ratio of the krill was based on measurements from MOCNESS catches in 2010 and each $TS(f, L_A)$ was averaged over a Gaussian distribution of orientation angles ($N(0, 30)$) (Wiebe et al. 2013).

The DWBA-predicted $\sigma_{bs}(f_{120}, L_A)$ were used to calculate a length-weighted mean σ_{bs} at 120 kHz ($\overline{\sigma_{bs}}(f_{120})$). The distribution densities ($d_L(L_A)$) from the combined acoustic-length density distribution were used to weight $\sigma_{bs}(f_{120}, L_A)$,

$$\overline{\sigma_{bs}}(f_{120}) = \sum_{L_A=1}^{50 \text{ mm}} d_L(L_A) * \sigma_{bs}(f_{120}, L_A), \quad (1)$$

where the summation to 50 mm encompasses the full range of measured krill lengths. Next, $\overline{\sigma_{bs}}(f_{120})$ was used to scale volume backscatter ($S_v(f_{120})$, dB re m² m⁻³) and areal backscatter (s_A , m² nmi²) (MacLennan et al. 2002) at 120 kHz to numeric density (D_v , # m⁻³) and (D_A , # m⁻²), respectively.

Acoustic data

Data acquisition, postprocessing, and analyses are detailed in Jech and Michaels (2006), Overholtz et al. (2006), and Jech and Sullivan (2014). During the survey years, two different vessels (the NOAA ship Delaware II from 1999 to 2011 and the NOAA ship Pisces in 2012) and echosounders (the Simrad EK500 [Bodholt et al. 1989] from 1999 to 2008 and the EK60 [Andersen 2001] from 2009 to 2012) were used. In addition, different acoustic frequencies were collected: 12-kHz data were collected from 1999 to 2001, 18-kHz data from 2002 to 2012, 38-kHz and 120-kHz data from 1999 to 2012, and 70-kHz and 200-kHz data in 2012. The acoustic data were calibrated prior to each survey using the standard sphere method (Foote et al. 1987; Demer 2015). Data from the surface to 10 m (to account for transducer depths on the keel or retractable keel, and to eliminate transducer “ring down,” surface bubbles, and surface noise) and deeper than 0.5 m above the seabed echo were eliminated from analysis. Data were vertically integrated into 1-m range bins so that the range resolution between the EK500 and EK60 was equivalent. A “spike reduction” filter (Jech 2014) (aka, “impulse noise” [Ryan et al. 2015]) was applied to minimize spurious strong echoes in the 120-kHz data, and a 3×3 weighted kernel, with maximum weight applied to the center element (Jech 2011) was applied to smooth the data. Although the NOAA ship Pisces collected 18 kHz, 38 kHz, 120 kHz, and 200 kHz data, only the 18 (f_1), 38 (f_2), and 120 (f_3) kHz data were used to maintain consistency among years. Processing was conducted in Echoview (v8+, Echoview Software Pty Ltd, Hobart, Tasmania) and data analysis was conducted in R version 3.43 (R-Core-Team 2015).

After these processing steps had been applied, four different classification methods were used to classify $S_v(f)$ as krill: the multifrequency single-beam index (MFSBI) (Jech and Michaels 2006; Wall et al. 2016), “dB-differencing” (Madureira

et al. 1993; Kang et al. 2002), the multifrequency index (MFI) (Trenkel and Berger 2013), and the Z-score (De Robertis et al. 2010; Ressler et al. 2012). The S_v bins classified as krill were mapped to the 120-kHz data, where the “krill” bins retained their S_v value and all other bins were set to -999 dB (i.e., zero) (see Supporting Information Fig. S1). The 120-kHz classified data were used to calculate areal backscatter (s_A , $m^2 \text{ nmi}^2$) (MacLennan et al. 2002) throughout the water column at 0.5 nmi horizontal intervals. These s_A values from only the parallel transects were used to estimate krill biomass for each year and survey.

Acoustic backscatter classification

MFSBI

Previous efforts at classifying these acoustic data have focused on separating backscatter by Atlantic herring (*C. harengus*) from other backscatter (Jech and Michaels 2006; Wall et al. 2016) and the MFSBI method has been used to generate herring abundance and biomass estimates from 1999 to 2012 (Jech and Sullivan 2014). As applied to Atlantic herring, the MFSBI method relies on a $S_v(f)$ threshold that minimizes non-herring backscatter and removes much of the backscatter by macrozooplankton. Although krill were not targeted by net sampling during the surveys, the cod-end liner of the mid-water trawl was $\frac{1}{4}$ inch-diameter (0.64 cm) hexagonal knotless nylon material, and appeared to capture krill with some correspondence to that observed in the acoustic data during the surveys. Because of the MFSBI's generic applicability and visual correspondence to krill catches in midwater trawl catches during the Atlantic herring surveys, the MFSBI method and midwater trawl catch data were used to provide initial data sets (i.e., “training” sets) for the Z-score method and comparison to the other three classification methods.

The S_v threshold was set to -66 dB for all frequencies, and each echogram bin where the 12/18-, 38-, and 120-kHz S_v values were greater than this threshold were coded with a “1,” “3,” “5,” respectively (Jech and Michaels 2006; Wall et al. 2016). The coded bins were summed across frequencies and bins that were coded as “5” (i.e., only 120-kHz S_v was greater than -66 dB) were classified as krill.

dB-differencing

The “dB-differencing” method (aka “delta mean volume backscattering strength” or ΔMVBS) relies on the difference in acoustic scattering response by krill between pairs of frequencies (Madureira et al. 1993; Kang et al. 2002; Lawson et al. 2008). Krill are considered a fluid-like scatterer (Stanton et al. 1998; Lawson et al. 2006; Lavery et al. 2007), and a dominant feature of their frequency response is that the Rayleigh scattering region spans the frequency range from < 1 kHz up to about 120 kHz. This feature has been exploited by comparing S_v and TS at “low” and “high” frequencies, where the most common pair is 38 kHz and 120 kHz, to classify acoustic backscatter to krill (e.g., Lawson et al. 2008; Reiss et al. 2008). For

this paper, echogram bins with a $\Delta S_v(f_{120} - f_{38})$ of 8 dB or greater (e.g., Madureira et al. 1993) were classified as krill. An initial $S_v(f)$ threshold of -100 dB was set for all frequencies so that all bins had a measured S_v value, whereas the final 120-kHz echogram had a threshold of -80 dB. The initial low threshold is needed so that all pairs of echogram bins (i.e., each corresponding bin between the 38-kHz and 120-kHz data) have measured values, rather than either frequency being arbitrarily set to -999 dB. The -80 dB threshold is similar to other surveys of euphausiids (Lascara et al. 1999; Kornelissen and Ona 2003; Lawson et al. 2008).

MFI

The multifrequency indicator (MFI) was developed to classify multifrequency echosounder data based on the acoustic scattering properties of organisms ranging from copepods to euphausiids to bubbles to fish with or without gas-filled swimbladders (Trenkel and Berger 2013). The basis of this method is that over many years of collecting and analyzing multifrequency acoustic data, four major groups of scattering types (fish with gas-filled swimbladders, fish without gas-filled swimbladders, fluidlike zooplankton, and small gas-bearing organisms) have been found to have backscattering characteristics that are consistently predictable over many environmental and biological conditions and are different among these groups so that classification to these scattering types is consistently robust. This method was based on ecological analyses where pairwise comparisons of species abundances are made to develop overall species diversity indices. By analogy, the S_v frequency responses are used to develop an index, and this index is compared to trawl and net catches to validate the index. For this paper, frequency-independent parameters were set identical to Trenkel and Berger (2013), and the bins classified as krill were those with MFI values ≥ 0.7 and ≤ 0.83 . An initial $S_v(f)$ threshold of -100 dB was set for all frequencies so that all bins had a S_v value, whereas the final 120-kHz echogram had a threshold of -80 dB.

Z-score

The Z-score method expands “dB-differencing” by quantifying and combining pairwise comparisons among all available frequencies into a single metric (De Robertis et al. 2010; Ressler et al. 2012). The Z-score method compares the frequency response for each frequency pair in each echogram bin (b) to a reference mean normalized by the reference standard deviation for each frequency pair (k) and taxon (m)

$$Z(k, m)_b = \Delta S_v(k)_b - \frac{\overline{\Delta S_v(k, m)}}{\text{SD}(\Delta S_v(k, m))}. \quad (2)$$

For each echogram bin, the mean Z-score ($\overline{Z(m)}_b$) for all three frequency pairs was computed by summing the absolute value of each $Z(k, m)_b$ and dividing by three.

Table 1. Empirically derived Z-score means and standard deviations (in parentheses) for each year, and the DWBA-assisted Z-score means and standard deviations that were applied among all years are given in the last row.

Year	$\overline{S_v}(f_{12}-f_{38})$	$\overline{S_v}(f_{18}-f_{38})$	$\overline{S_v}(f_{120}-f_{38})$	$\overline{S_v}(f_{120}-f_{12})$	$\overline{S_v}(f_{120}-f_{18})$
1999	-5.3 (3.0)	---	13.8 (2.3)	19.2 (3.1)	---
2000	-6.4 (5.7)	---	8.1 (3.7)	14.8 (5.4)	---
2001	-12.3 (3.9)	---	12.9 (3.3)	25.2 (4.2)	---
2002	---	-2.3 (3.3)	10.7 (3.0)	---	13 (4.7)
2003	---	-7.6 (4.9)	10.2 (2.7)	---	17.9 (5.3)
2004	---	-6.1 (4.2)	14.6 (3.0)	---	20.6 (5.1)
2005	---	-8.2 (3.2)	9.8 (2.2)	---	18.0 (3.6)
2006	---	-6.5 (4.1)	12.4 (2.9)	---	18.8 (5.5)
2007	---	-8.2 (4.2)	11.6 (3.4)	---	19.8 (6.5)
2008	---	-4.7 (3.9)	10.5 (3.1)	---	15.2 (4.5)
2009	---	-1.9 (4.1)	12.9 (2.9)	---	14.9 (4.4)
2010	---	-1.4 (3.1)	10.8 (2.3)	---	12.4 (3.6)
2011	---	-4.1 (3.6)	12.6 (2.4)	---	17.0 (3.9)
2012	---	-4.1 (5.3)	11.8 (4.0)	---	15.9 (6.1)
DWBA	-18.1 (0.85)	-11.4 (0.65)	14.3 (1.67)	32.4 (2.49)	25.8 (2.32)

An initial set of bins classified as krill was generated with $\overline{Z}(m)_b \leq 2$ and this set was mapped to the 120-kHz S_v data. As recommended by De Robertis et al. (2010), a second “pass” through the data was completed. The spatial resolution of the initially classified 120-kHz S_v data was reduced where the 1 ping $\times$ 1 m Z-scores and S_v were averaged into bins of 0.5 nmi horizontal by 10 m vertical. Reduced-resolution bins with $\overline{Z}(m)_b \leq 1.5$ were then classified as krill and this second set of classified bins was mapped to the 120-kHz S_v echogram with the reduced resolution for calculations of s_A .

Subsets of data were selected each year for initial evaluation and to provide the frequency-dependent reference means and standard deviations used by the Z-score method. The subsets were selected by visually comparing the MFSBI classification to the original data echograms, and, if possible, in close spatial and temporal proximity to trawl catches with high catches of krill. These subsets were assumed to constitute bins with

species composition of 100% krill, i.e., taxon m = krill. Each year’s subset of $S_v(f)$ data were used to generate the reference mean ($\overline{\Delta S_v}(k, m)$) and standard deviation (SD) of $\Delta S_v(k, m)$ for frequency pair (k) (12 and 38 kHz, 18 and 38 kHz, 12 and 120 kHz, 18 and 120 kHz, and 38 and 120 kHz) (Table 1)

$$\overline{\Delta S_v}(k, m) = \frac{1}{N_b} \sum_{b=1}^{N_b} \Delta S_v(k, m)_b \quad (3)$$

$$SD(\Delta S_v(k, m)) = SD\left(\Delta S_v(k, m)_{|b=1}^{N_b}\right) \quad (4)$$

The reference mean and standard deviation were computed in the $\log_{10}$ (i.e., “dB”) domain for each year and applied to the corresponding year. An initial $S_v(f)$ threshold of -100 dB was set for all frequencies so that all bins had a S_v value, whereas the final 120-kHz echogram had a threshold of -80 dB.

DWBA-assisted Z-score

The Z-score classification method (De Robertis et al. 2010; Ressler et al. 2012) uses empirical acoustic measurements to determine the reference parameters mean and standard deviation. Because of the potential for contamination by other scatterers in the data sets used to generate the empirically derived reference values, we developed DWBA-based parameter values using the predicted target strengths and length distributions. One thousand Gaussian distributed acoustic lengths were generated based on the mean and standard deviation of the acoustic lengths ($N(\overline{L_A}, SD(L_A))$). Each length was mapped to its predicted $TS(f, L_A)$ for each frequency. For this analysis, we assume the hypothetical sampling volume contains only krill and so $S_v(f)$ is directly represented by TS, and because the Z-score parameters are derived from and the classification is applied to the $S_v(f)$ data, the means and standard deviations are calculated using TS (i.e., log-10 based dB values). In other words, TS and S_v are used interchangeably when calculating the mean and SD of $\Delta S_v(f_1 - f_2)$ for frequency pairs: $S_v(12) - S_v(38)$, $S_v(18) - S_v(38)$, $S_v(120) - S_v(38)$, $S_v(120) - S_v(12)$, and $S_v(120) - S_v(18)$:

$$\overline{\Delta S_v(f_1 - f_2)} = \frac{1}{1000} \sum_{i=1}^{1000} (TS(f_1, L_{A_i}) - TS(f_2, L_{A_i})), \quad (5)$$

$$SD(\Delta S_v) = SD\left(TS(f_1 - L_{A_i}) - TS(f_2 - L_{A_i}) \Big|_{i=1}^{1000}\right) \quad (6)$$

By indexing over acoustic length, the length is consistent for each frequency pair (i.e., the difference is not done over random pairs of $TS(f, L_A)$) and is representative of the overlapping sampling volumes insonified by the acoustic beams. These DWBA-based parameter values were used in the DWBA-assisted Z-score method.

Biomass estimation

Acoustic survey

The mean s_A for the entire common survey region for each year was obtained by calculating the mean s_A for each individual parallel transect within the common survey area and taking the mean of those transect means. The standard error (SE) of s_A for the common region was calculated by dividing the standard deviation of the transect means by the square root of the number of transects. The mean areal numerical density for the common survey region ($\overline{D_{A,CR}}$, # nmi⁻²) was calculated by:

$$\overline{D_{A,CR}} = \frac{\overline{s_{A,CR}}}{4\pi\overline{\sigma_{bs}}(f_{120})}. \quad (7)$$

Multiplying $\overline{D_{A,CR}}$ by the area of the common region ($A_{CR} = 9279$ nmi²) gave the total abundance (P , #). Total biomass (B_{CR} , mt) was calculated by multiplying abundance (P) by the mean weight ($\overline{W}$) (weighted by the length distribution

($d_L(L_A)$)) and scaling milligrams to metric tonnes (factor of 10⁹). Standard error of biomass was calculated by multiplying the standard error of the s_A values (SE_{CR}) by the constant (C) that was used to convert s_A to biomass (B_{CR}):

$$C = \frac{\overline{W}^* A_{CR}}{4\pi\overline{\sigma_{bs}}(f_{120})10^9}. \quad (8)$$

The standard error of biomass is then $SE_B = C * SE_{s_A}$. Because krill biomass is not known and other biomass estimates do not exist with which to compare, two metrics of performance were developed as indicators to rank the classification methods: annual total biomass was divided by its standard error (analogous to a signal to noise ratio), and the mean, over all years, of this metric for each classification method.

Independent net samples

Areal density (ρ_A , # m⁻²) and acoustic length estimates from MOCNESS samples were used to estimate krill biomass for comparison to the acoustic survey-derived biomass estimates. The mean acoustic length from each sample was converted to SL1, and that length was used to calculate the mean weight using McQuinn et al. (2015) length-weight relationship for each sample. Sample areal density was converted to biomass, and then scaled (1852 m nmi⁻¹) to the survey area (9279 nmi²) to give total biomass (metric tonnes) for the survey area for each sample.

Annual mean areal density (# 10 m⁻²) (where sample volume was scaled by the vertical extent of each tow) from bongo samples collected during the NEFSC Ecosystem Monitoring Surveys (hereafter, "Ecomon") (Kane 2007; Lowe et al. 2018) and the mean SL1 from the combined (1999–2012) length distribution (Fig. 2E) were used to estimate krill biomass. Areal density was converted to biomass using McQuinn et al. (2015) length-weight relationship and scaled to the survey area to estimate total biomass for the survey area for each year.

Results

Krill lengths

Krill length measurements from the study regions were obtained from the NEFSC midwater trawl (MWT) and Methot catch data in 1997 and 1998, stomach samples from the NEFSC bottom trawl survey during 2002–2012, and MOCNESS tows from an experimental cruise in 2010 (Wiebe et al. 2013) (Figs. 1–2). Krill were not identified to species for the MWT, Methot, and stomach samples, but analysis of a subset of the MOCNESS samples (tow MOC1 and 2) indicated that *M. norvegica* was numerically dominant at depths below 75 m (ranging among nets from 84% to 100% of catch). From the surface to 75 m, the community composition was more variable, with mostly small *M. norvegica* making up 23–96% of catches, and a variety of smaller krill species also present, particularly *Euphausia krohnii* and *Nematoscelis megalops*. Although the Methot and MWT tows from 1997 were conducted outside

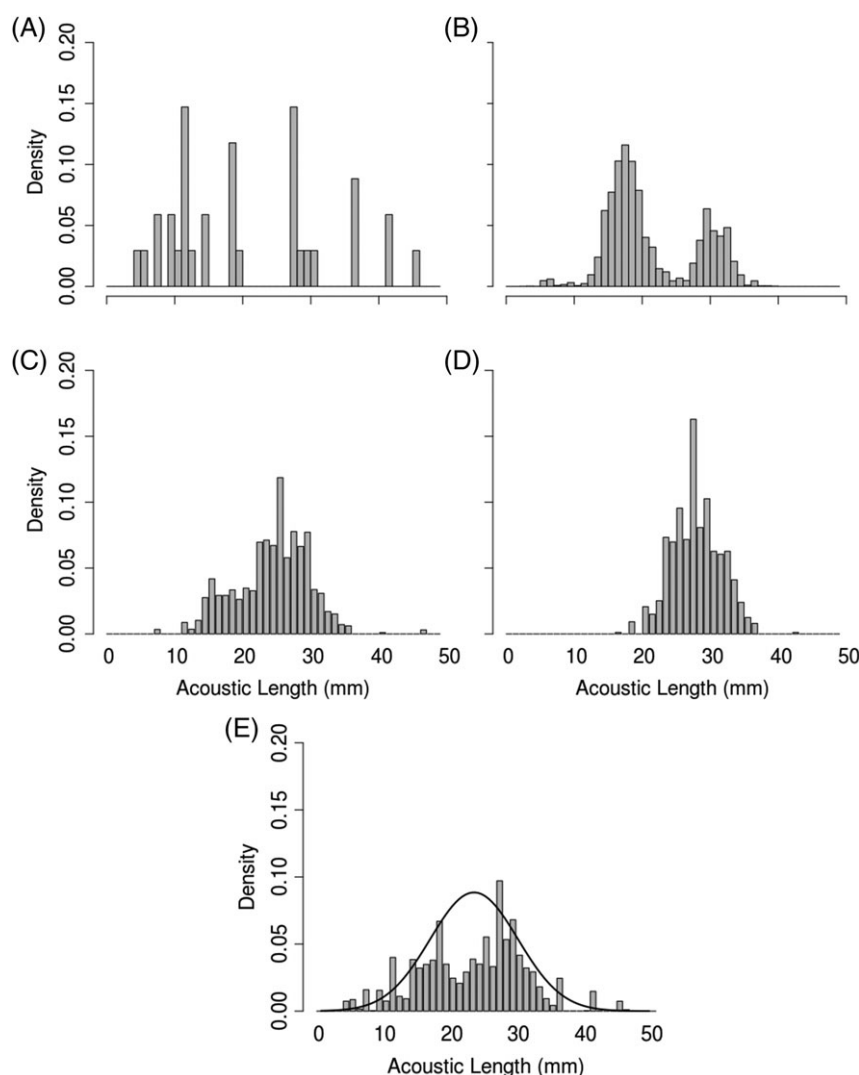


Fig. 2. Krill acoustic-length distributions from (A) finfish stomach samples and (B) multiple opening/closing net and environmental sampling system (MOCNESS), (C) Methot and (D) midwater trawl catches. The combined length distribution and Gaussian curve (mean = 23 mm, standard deviation = 7.9 mm) are shown in panel (E).

of the Georges Bank region (Fig. 1), earlier net-catch data suggest *M. norvegica* are the dominant species of krill in that area as well (Greene et al. 1992; Warren et al. 2003) and that the lengths can be combined for the different areas. The combined acoustic lengths for all MWT catches showed a unimodal distribution (Fig. 2D) with a mean and standard deviation of 28 mm and 3.6 mm, respectively. The combined acoustic lengths for all Methot catches showed a bimodal distribution (Fig. 2C) with modes at ~ 15 mm and 25 mm, and overall mean and standard deviation of 24 mm and 5.4 mm, respectively.

Thirty-four krill lengths with digestive rank of “fresh” or “partial” (meaning the krill was entirely or mostly intact) were recorded from stomach samples from fish collected during the NEFSC acoustic/midwater trawl surveys in 2002, 2008, 2010, 2011, and 2012. The SL1 measurements were converted to acoustic length and the lengths spanned from 4 mm to

49 mm (Fig. 2A) with a mean and standard deviation of 20 mm and 11.6 mm, respectively.

The combined length distribution for MOCNESS samples showed a strongly bimodal distribution with modes at ca. 18 mm and 30 mm (Fig. 2B) and overall mean and standard deviation of 21 mm and 6.7 mm, respectively. The acoustic-length distribution of all length data combined was multimodal, with the two strongest modes preserved from the MWT, Methot, and MOC length distributions, and a smaller mode at ~ 10 mm that reflects the shorter prey lengths (Fig. 2E). The mean and standard deviation of the combined data were 23 mm and 7.9 mm, respectively.

DWBA predicted backscatter

The DWBA-predicted $TS(f, AL)$ (Fig. 3A) were used to calculate the $\Delta TS(f, AL)$ curves (Fig. 3B) that were used to generate

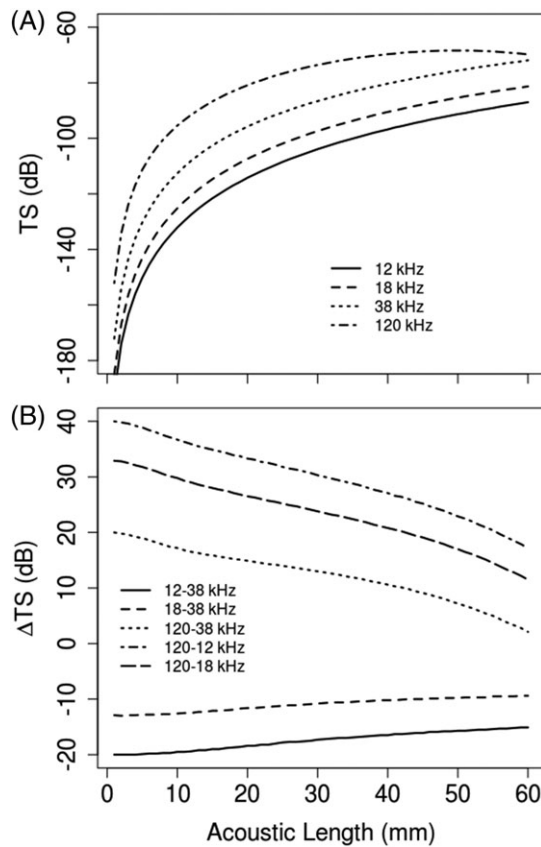


Fig. 3. Distorted wave Born approximation (DWBA) predicted target strength (TS dB re m^2) as a function of krill acoustic length at 12 kHz, 18 kHz, 38 kHz, and 120 kHz (panel A), and the difference in TS for each combination of frequency pairs as a function of krill acoustic length (panel B).

the $\Delta S_v(f)$ means and standard deviations for the DWBA-assisted Z-scores (Table 1). The curves show the typical response of a fluid-like scatterer with a transition from Rayleigh scattering region at short lengths (i.e., low ka , where k is the acoustic wave number ($k = 2\pi/\lambda$, where λ is the acoustic wavelength and a is the radius of the krill) to geometric scattering at longer lengths. For the krill lengths in the Gulf of Maine and Georges Bank regions, the TS curve at 120 kHz leveled off at about 40 mm length, whereas the curves at the other frequencies were still in the transition region up to the longest acoustic lengths encountered (48 mm).

The slopes of the $\Delta S_v(k)$ curves (Fig. 3B) directly affected the DWBA-assisted Z-score standard deviations, where the steeper-sloped curves resulted in greater standard deviations (last row in Table 1; Fig. 4). The $\Delta S_v(f_{12} - f_{38})$ and $\Delta S_v(f_{18} - f_{38})$ were relatively flat with less than 5 dB change as compared to the curves with 120 kHz ($\Delta S_v(f_{120} - f_{12})$, $\Delta S_v(f_{120} - f_{18})$, and $\Delta S_v(f_{120} - f_{38})$) where the $\Delta S_v(k)$ changed 18–23 dB over the range of acoustic lengths (Fig. 3B). This resulted in smaller standard deviations for the $\Delta S_v(f_{12} - f_{38})$ and $\Delta S_v(f_{18} - f_{38})$ Z-score standard deviations (Fig. 4A) relative to those standard

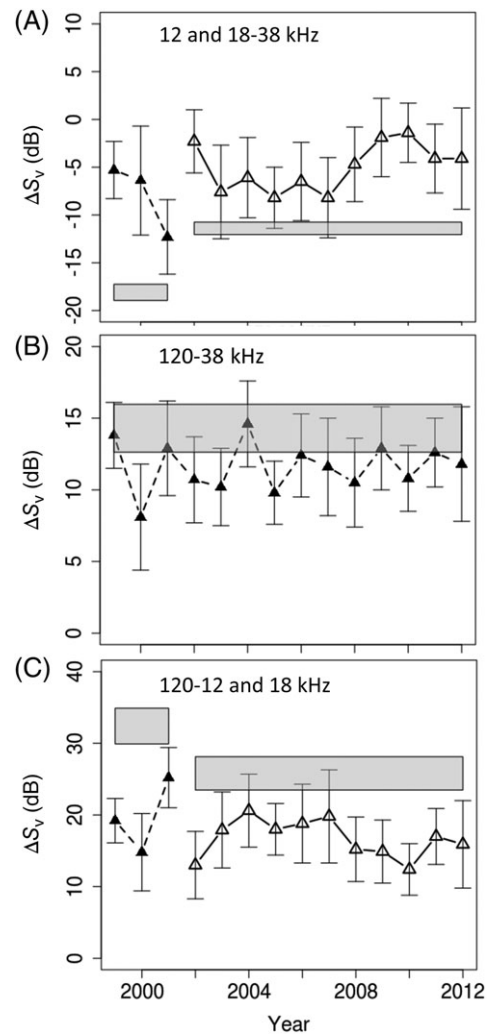


Fig. 4. Empirically based (derived from the MFSBI classification method) reference means (solid triangles denote 12 kHz and open triangles denote 18 kHz in panels A and C) and standard deviations (error bars) of $\Delta S_v(f_1 - f_2)$ used by the Z-score method, and the range of DWBA-assisted reference means and standard deviations (shaded rectangles) for frequency pairs (A) 12 kHz, 18 kHz, and 38 kHz, (B) 120 kHz and 38 kHz, and (C) 12 kHz, 18 kHz, and 120 kHz from 1999 to 2012.

deviations that were based on 120 kHz ($\Delta S_v(f_{120} - f_{12})$, $\Delta S_v(f_{120} - f_{18})$, and $\Delta S_v(f_{120} - f_{38})$) (Table 1; Fig. 4B,C).

Acoustic classification

The empirically based reference means and standard deviations of $\Delta S_v(k)$ for each frequency pair used by the Z-score classification method (Table 1) were calculated from regions classified as krill by the MFSBI method for each year, and were applied to each respective year. The DWBA-assisted means and standard deviations were applied equally among all years for each frequency pair (Table 1). Because we had acoustic data for each year from which to derive the empirical reference values, we were able to use annual reference values; whereas because of the lack of sufficient length data for many years,

we needed to combine the length distributions among years and derive a common set of DWBA-assisted reference values that were applied equally among all years. Comparisons of the empirically based and DWBA-assisted means $\overline{\Delta S_v}(k)$ and standard deviations showed no overlap for the frequency pairs with 12 kHz ($\Delta S_v(f_{12} - f_{38})$ and $\Delta S_v(f_{120} - f_{12})$) (Fig. 4A), overlap in 3 yr during 2003–2007 for the frequency pairs with 18 kHz ($\Delta S_v(f_{18} - f_{38})$ and $\Delta S_v(f_{120} - f_{18})$) (Fig. 4A,C), and strong overlap in all years except 2000 and 2005 for the frequency pair $\Delta S_v(f_{120} - f_{38})$ (Fig. 4; Table 1).

The DWBA-assisted Z-score standard deviations were smaller than the empirically based standard deviations (Table 1). The DWBA-assisted standard deviations for $\Delta S_v(f_{12} - f_{38})$ and $\Delta S_v(f_{18} - f_{38})$ were nearly an order of magnitude smaller, and the DWBA-assisted standard deviations for the other frequency pairs were about half those of the empirically based standard deviations. For this reason, the acceptance criterion for the DWBA-assisted Z-score classification was set to 6 rather than the 1.5 used for the empirically based Z-score classification.

Biomass estimates

Biomass estimates were derived from the acoustic surveys, MOCNESS tows, and annual density estimates from the Ecomon Survey (Kane 2007). For the acoustic and Ecomon survey estimates, the overall mean acoustic length of 23.0 mm was converted to SL1 (25.3 mm), and that was used to estimate the mean weight of a krill at 0.143 g and the mean TS at -76.2 dB. The mean weight and TS were used to generate the annual biomass estimates from the acoustic surveys and from the Ecomon surveys. For the MOCNESS tows, acoustic length distributions were provided for each tow, so acoustic length was converted to SL1 and the mean SL1 for each tow was used to estimate a weight and then scaled to the survey area for four biomass estimates in 2010.

The spatial distribution of krill as classified from the acoustic data was patchy with aggregation lengths (distance along a transect) ranging from 0.5 nmi to 63 nmi (Fig. 5). The lateral extents (i.e., between transects) of the patches are not known. Transect spacing was nominally 10 nmi over the years, so with along-transect lengths of 10s of nautical miles and given the often similar along-transect distribution of biomass on adjacent transects, it is likely that the same patches were observed on different transects.

With the exception of the year 2000, biomass estimates of Northern krill derived from the acoustic surveys in the Georges Bank region ranged from 56.3 to about 1331 1000 mt (Table 2; Fig. 6A). In 2000, the Z-score and MFI methods estimated the highest biomass at 7058 and 3053 1000 mt, respectively. Elevated (greater than the overall mean of 438 1000 mt, excluding biomass estimates in 2000) biomass was estimated by the dB-differencing, MFI, and Z-score classification methods in 2000, 2002–2004, 2006–2007, and 2012. When estimates from the year 2000 were excluded, the mean and standard

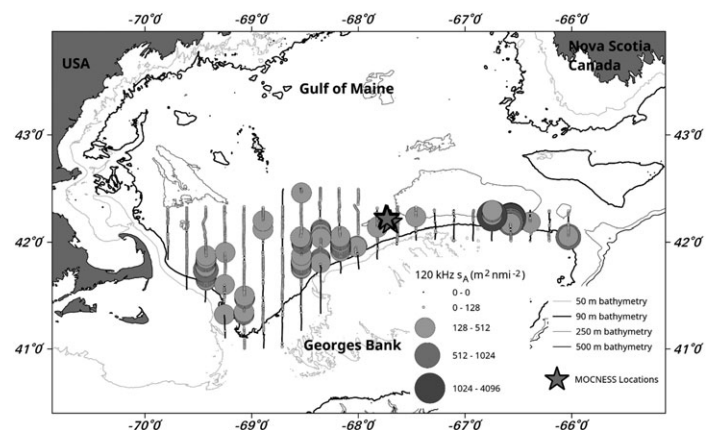


Fig. 5. 120-kHz areal backscatter (s_A $\text{m}^2 \text{nmi}^{-2}$) classified as krill by the DWBA-assisted classification method at 0.5-nmi intervals along the acoustic survey transects in 2010. Gray stars denote the multiple opening/closing net and environmental sampling system (MOCNESS) sampling locations in 2010.

error estimates for all methods were reduced (“exclude year” row in Table 2). The proportional change in mean biomass and standard error was greatest for the empirically based Z-score method, and least for the DWBA-assisted Z-score and dB-differencing methods.

The MFSBI classification method estimated overall lowest biomass, with the DWBA-assisted Z-score method estimating slightly more biomass than the MFSBI method over all years, but both of these methods estimated less than half of what the other three methods estimated (Table 2). The empirically based Z-score method predicted nearly three times the biomass than the MFSBI or DWBA-assisted Z-score methods. Dividing the annual biomass estimates by their respective standard errors highlighted years (2002, 2003–2004) where the variability in biomass estimates was high, and a trend of increasing variability relative to mean biomass starting in 2007 (Fig. 6B). High or low biomass to standard error ratios do not appear to be related to high or low biomass estimates. For example, lowest biomass was estimated in 2005 and 2011, yet the biomass to standard error ratios were high. The MFSBI and DWBA-assisted Z-score classification methods had the lowest overall biomass to standard error ratios (i.e., they had proportionally the highest variability), with the MFI, dB-differencing, and empirical-based Z-score having increasing ratios (Table 2).

The MOCNESS biomass estimates in 2010 were 839, 588, 167, and 3657 1000 mt for MOCNESS tows MOC1, MOC2, MOC3, and MOC4, respectively (Fig. 6A; Table 2). The biomass estimate from MOC2 was commensurate with the acoustic survey estimates, whereas the biomass estimate from MOC3 was lower and biomass estimates from MOCs 1 and 4 were greater than the acoustic estimates, with MOC4 estimating nearly 3-times more biomass than the acoustic survey. The annual biomass estimates from the Ecomon surveys were in general an order of magnitude lower than the acoustic survey and MOCNESS-based biomass estimates. Kane (2007),

Table 2. Annual biomass estimates ($\times 10^3$ mt) and standard errors (values in italics) for the classification methods: multifrequency single beam index (MFSBI), dB-differencing (Δ dB), multifrequency index (MFI), Z-score, and DWBA-assisted Z-score (DWBA). “Mean biomass” is the mean, over all years, of the annual biomass estimates (with standard error values in italics), and “mean mt/SE” is the mean, over all years, of the annual biomass estimates divided by their standard errors. The biomass estimates derived from the NEFSC Ecomon surveys are under “Ecomon,” and those derived from the MOCNESS tows (#1–4 in the order they are presented) in 2010 are under “MOC.” Mean biomass was calculated using all years, and “exclude year” excluded the year 2000 from the acoustic methods and the year 2004 from the Ecomon estimates.

Year	MFSBI	Δ dB	MFI	Z-score	DWBAZ-score	Ecomon	MOC
1999	129.76 <i>42.67</i>	547.63 <i>121.84</i>	809.79 <i>239.29</i>	212.01 <i>61.14</i>	56.3 <i>22.87</i>	256.44	
2000	1151.13 <i>341.77</i>	1506.54 <i>345.29</i>	3053.54 <i>865.21</i>	7058.06 <i>1338.07</i>	606.13 <i>319.78</i>	179.80	
2001	207.18 <i>68.18</i>	477.25 <i>98.53</i>	467.58 <i>111.73</i>	285.47 <i>78.30</i>	214.65 <i>70.38</i>	48.79	
2002	212.45 <i>56.30</i>	817.27 <i>120.52</i>	638.68 <i>106.45</i>	1331.47 <i>160.99</i>	630.77 <i>188.70</i>	87.50	
2003	310.98 <i>169.79</i>	760.97 <i>247.20</i>	708.18 <i>222.13</i>	987.50 <i>294.27</i>	560.39 <i>165.39</i>	81.90	
2004	306.59 <i>62.02</i>	1070.63 <i>129.76</i>	798.35 <i>104.69</i>	633.40 <i>87.53</i>	429.75 <i>72.58</i>	0.00	
2005	86.65 <i>25.07</i>	379.16 <i>47.07</i>	324.62 <i>46.19</i>	316.70 <i>54.10</i>	225.21 <i>40.03</i>	4.37	
2006	238.41 <i>86.65</i>	501.89 <i>109.53</i>	501.89 <i>111.73</i>	627.69 <i>138.56</i>	485.17 <i>100.73</i>	22.63	
2007	362.89 <i>173.31</i>	864.33 <i>267.44</i>	678.27 <i>248.96</i>	738.97 <i>245.00</i>	297.79 <i>95.89</i>	19.37	
2008	82.25 <i>29.31</i>	384.00 <i>84.01</i>	340.02 <i>84.89</i>	545.43 <i>97.65</i>	283.71 <i>75.22</i>	9.35	
2009	202.34 <i>53.22</i>	503.64 <i>88.85</i>	328.14 <i>71.26</i>	508.92 <i>106.45</i>	173.75 <i>50.14</i>	33.68	
2010	171.55 <i>62.46</i>	375.64 <i>86.65</i>	256.00 <i>70.82</i>	430.19 <i>90.17</i>	146.04 <i>50.14</i>	108.49	839.30 588.39 167.35 3657.36
2011	92.81 <i>17.15</i>	321.54 <i>43.11</i>	211.13 <i>32.11</i>	203.22 <i>36.51</i>	83.57 <i>21.11</i>	31.53	
2012	307.03 <i>83.05</i>	606.57 <i>118.59</i>	488.69 <i>103.81</i>	871.81 <i>139.70</i>	310.54 <i>69.23</i>	35.47	
Mean biomass	275.86 <i>71.54</i>	651.22 <i>88.01</i>	686.06 <i>189.64</i>	1053.63 <i>469.59</i>	321.70 <i>51.07</i>	65.67 <i>19.70</i>	1313.13 793.62
Exclude year	208.53 <i>26.11</i>	585.42 <i>63.13</i>	503.95 <i>57.13</i>	591.75 <i>91.52</i>	299.82 <i>49.84</i>	70.72 <i>20.56</i>	
Mean mt/SE	3.34	5.34	4.69	5.11	3.73		

using catch data from the Ecomon surveys, estimates a mean density of 474 ($\# 100 \text{ m}^{-3}$) Euphausiacea for Georges Bank over the time period from 1977 to 2004, which translates to a biomass estimate of 431 1000 mt of krill.

Discussion

The biomass estimates derived from the acoustic/midwater trawl survey represent an initial effort to quantify the spatial

distribution and biomass of krill in the Georges Bank region using acoustical methods. Acoustical methods are used to monitor and assess Antarctic krill (*Euphausia superba*) (Demer 2004; Reiss et al. 2008) on a routine basis, and have been applied to “multipurpose” multifrequency acoustic data collected during other surveys (Ressler et al. 2012; McQuinn et al. 2015), but to our knowledge are not used for routine population assessments elsewhere. Acoustically, krill are a fluid-like scatterer (Stanton et al. 1998; Demer and Conti 2003a; Lavery et al. 2007) and for

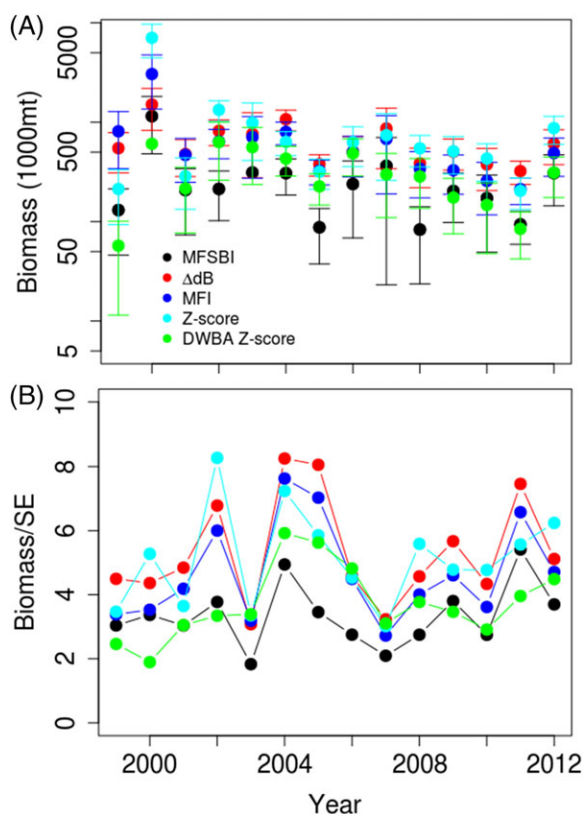


Fig. 6. Biomass (1000 mt) estimates of krill based on the acoustic classification methods (solid colored circles) from 1999 to 2012 (panel A). Panel B shows the biomass estimate to standard error ratio (solid circles with color corresponding to panel A) of the acoustic classification methods from 1999 to 2012.

the sizes of krill found in the Georges Bank region, the acoustic scattering properties allow straight-forward detection and classification using multiple acoustic frequencies (Mair et al. 2005; Korneliussen et al. 2016). Furthermore, in the present study region along the flank of Georges Bank krill constitute the dominant zooplankton scatterer, accounting for > 90% of estimated backscattering based on MOCNESS samples (at 43 kHz, 120 kHz, and 200 kHz; Wiebe et al. 2013); this is in contrast to other regions, including elsewhere in the Gulf of Maine, where scatterers other than krill (e.g., siphonophores) dominate scattering and complicate acoustic identification of krill (Lavery et al. 2007).

Agreement in biomass estimates (i.e., orders of magnitude differences) among the different acoustical methods is to be expected given their basis on the same fundamental scattering property of krill. The classification methods used here utilize the dominant scattering feature of krill which is defined by lower intensity scattering at 12 kHz, 18 kHz, and 38 kHz than at 120 kHz, i.e., spans the transition from Rayleigh to geometric scattering, and that is presumably why the biomass estimates are fairly consistent among the methods. Spanning that transition zone is critical for the classification methods, but using “lower” (e.g., 12 kHz or 18 kHz) frequencies for weakly

scattering fluid-like scatterers is challenging. Because the target strength by individual krill is lower at these frequencies than at higher frequencies, there is increased potential for a negative bias in detectability during data acquisition and for removing backscatter during processing and analysis — i.e., backscattering by krill can be lost in electrical and acoustical noise at the lower frequencies. The 38 kHz and 120 kHz data were more aligned with the theoretical predictions than were those with the 12 kHz or 18 kHz data, suggesting that adding “low” frequencies may not provide advantages to classification of krill. However, each method is different in how it implements its classification scheme and ultimately the derivation of the final biomass estimates.

The MFSBI and DWBA-assisted classification methods generated the lowest biomass estimates—nearly half the biomass of the other three methods and the MOCNESS-based biomass estimates. The MFSBI was originally developed to highlight acoustic scattering by swimbladder-bearing fish (Jech and Michaels 2006), but the overall method does allow for classification of other scattering features, such as those from krill (Wall et al. 2016). The primary restriction of MFSBI is the choice of S_v threshold. We used a -66 dB S_v threshold, which for a krill with a mean TS of -76.2 dB means that the minimum density of detection will be approximately 10.5 krill per cubic meter. In addition, when krill densities are very high, S_v at 18 kHz or 38 kHz can be greater than -66 dB, resulting in the exclusion of those echogram cells in the classification scheme (i.e., cells not classified as krill). For example, at 38 kHz, the DWBA-predicted mean TS of -88.3 dB means that krill densities greater than about 170 per m^3 will not be classified as krill, and these high-density patches will not contribute to the overall biomass estimate. Thus, the MFSBI has upper and lower acceptance thresholds for classifying krill. While the lower threshold is not an unreasonably low detection threshold (e.g., Warren et al. 2003), the upper threshold probably does limit the higher density S_v cells, and the combination of these two thresholds does seem to have the effect of lowered biomass estimates for the MFSBI method relative to the other acoustic methods and net-based estimates.

The empirically based Z-score classification method, on the other hand, had the highest biomass estimates of all acoustic methods and net-based estimates, and in some years (e.g., 2000) these estimates seemed unreasonably high. The reference means and standard deviations were generated from regions classified as krill by the MFSBI method, and a comparison of the empirically based reference means and standard deviations to those based on the DWBA model predictions suggested that the training sets, especially at 12 kHz and 18 kHz, were contaminated by other scatterers such as small fish and salp species that are found in the area (e.g., Warren et al. 2003; Lavery et al. 2007; Wiebe et al. 2013), which potentially led to overestimation of krill biomass. The year 2000 seems to be an interesting year for the Georges Bank region, where acoustic data highlighted a strong change in

acoustic frequency response (Jech and Michaels 2006), suggesting an increase in fluid-like scatterers, such as salps or the gelatinous portion of siphonophores that affected the classification methods used here. Jech and Michaels (2006) hypothesized the change in frequency response was due to an increase in siphonophores in the near-surface layer. Siphonophores are gas-bearing animals, either solitary or aggregated, that are mostly gelatinous (Mackie et al. 1988). Acoustically, however, the echo from a siphonophore is dominated by the gas-bubble (Warren et al. 2001; Benfield et al. 2003; Trevorrow et al. 2005). In situ (Benfield et al. 2003) and ex situ (Stanton et al. 1994) measurements of physonect siphonulae showed a resonance feature that ranged from about 40–200 kHz depending on size of the bubble and depth. While the Z-score, DWBA-assisted Z-score, and MFI classification methods should filter out resonance features (the krill frequency response has no sharp resonance peak), the MFSBI and dB-differencing methods could accept backscatter by siphonulae as krill under fairly specific conditions (i.e., the bubble radius and depth correspond so that the frequency response is dominant at 120 kHz). The presence of a resonant feature is difficult to discern with multifrequency narrowband data, but in the future, wideband data (e.g., Jech et al. 2017) should have the spectral resolution to positively identify resonant features and hence eliminate that backscatter being classified as krill.

The DWBA-assisted Z-score classification method was developed to compensate for the potential contamination of the training sets (e.g., year 2000) and to determine the reference means and standard deviations of $\Delta S_v(k)$ for the empirically based Z-score method. This method resulted in “conservative” biomass estimates that were comparable to the MFSBI method and 2–4 times lower than the other acoustic methods and MOCNESS-based estimates. Likely as a consequence of avoiding contamination, the DWBA-assisted Z-score method was not affected as much as the other methods were in 2000. The DWBA-assisted Z-score biomass estimate in 2000 was comparable to estimates in other years and eliminating that biomass estimate had small to negligible impact on the mean and standard error of the overall mean. This suggests contamination is an issue in the data sets used to generate the empirical reference values. A formal sensitivity analysis was not done, but the Z-score method appeared to be most sensitive to the standard deviation parameter, where changes in the reference standard deviations had a larger effect on classification than did changes to the reference means. The smaller DWBA-predicted reference standard deviations required a larger acceptance criterion for the DWBA-assisted Z-score (acceptance criteria of 6) relative to the empirically based Z-score (acceptance criteria of 1.5). Because the DWBA-predicted reference means and standard deviations are restrictive, i.e., they assume 100% scattering by krill, the acceptance criterion is essentially superfluous when using model-based references.

Theoretical model predictions have been incorporated in classification algorithms to improve classification performance

(e.g., Demer and Conti 2003b; Korneliussen et al. 2016), and are an important component of interpreting acoustic data (Jech et al. 2015). Models can be used to predict acoustic frequency response for a full range of environmental conditions and animal material properties, which in turn can be used to bound target strength and volume backscatter measurements anticipated by the in situ species composition. The dB-differencing method has been in use for decades (Madureira et al. 1993; Trevorrow et al. 2005) and is regarded as the de facto standard for estimating krill abundance and biomass. However, because it is the de facto standard does not imply that it should remain static. The Z-score method advances the dB-differencing classification method by statistically combining multiple pairwise comparisons into a metric that can be used to quantify classification performance. By incorporating model-based acoustic frequency responses, our confidence in the classification is increased and the performance of the classification is potentially improved.

However, this improved confidence does come with a cost. The model must be properly parameterized and verified for the species and conditions encountered in situ. In the Georges Bank region, only limited measurements are available to inform model parameterization. For example, the magnitude and frequency response of TS are sensitive to orientation, but no observations have been made in the study region. Limited observations made elsewhere in the North Atlantic suggest a generally horizontal aspect (e.g., Kristensen and Dalen 1986) and following the approach of Wiebe et al. (2013) we therefore assumed a broad distribution of angles of orientation centered on horizontal. Further study of krill orientation and associated variability is highly warranted. Material properties of the dominant species of krill in the Georges Bank region, *M. norvegica*, are also not well characterized and without recent measurements (G. Lawson and S. Sakinan, unpubl. data), material properties of Antarctic krill (*E. superba*) or other euphausiid species from other parts of the world would have been used. Although the present study benefited from these recent measurements, *M. norvegica* acoustic material properties are known to vary seasonally (Køgelier et al. 1987; G. Lawson and S. Sakinan, unpubl. data) and may also vary in space as well as between years. Given the overall lack of information, fixed values for all model parameters were employed here to derive a single TS applied to all years; while this approach allowed us to focus on comparing among the acoustic methods, it does introduce uncertainty. Error in model parameters would affect acoustic estimates of biomass and further complicate the comparison between nets and acoustics. Unaccounted-for variability over the course of the time-series in particular model parameters could also affect the different acoustic methods differentially. Uncertainty in acoustic material properties, for instance, would affect the magnitude of scattering with consequences to the S_v threshold but not to the dB-differencing. In contrast, uncertainty in orientation, length, or length-to-width ratio would affect both the

magnitude of scattering and the shape of the frequency response, with consequences to dB-differencing. Exploring such possible consequences of spatial and temporal variability in model parameters lies beyond the scope of this initial work but will be an important component of future refinement and application of the present acoustic methods.

Comparisons of net-based (i.e., point) to acoustic-based (i.e., continuous) estimates should be made judiciously and extrapolation of point samples to larger areas should be done cautiously. Net tows targeted at dense layers can generate high biomass estimates, which are appropriate for that aggregation or layer, but will overestimate biomass when extrapolated to the water column or entire survey area, as exemplified by the concurrent echogram during MOC 4 (lower panel fig. 3 in Wiebe et al. [2013]). Oblique tows (e.g., MOCs 1 and 2) are more representative of the water column over which they sample, and therefore should provide better estimates of the vertically integrated abundance of organisms than targeted tows. In theory, these types of tows should be used to compare to the areal densities integrated over the entire water column derived from acoustic data. However, comparisons of MOCNESS-based biomass estimates from the oblique tows were 2–6 times greater than those derived from the acoustic data when extrapolated to the survey area, likely due to the fact that the MOCNESS sampling region was deliberately chosen for having overall high krill abundance in order to study krill behavior. s_A values from the two survey transects (conducted during 25–26 September 2010) surrounding the MOCNESS stations (conducted 24 and 26 September 2010) ranged from 0 $\text{m}^2 \text{ nmi}^{-2}$ to 160 $\text{m}^2 \text{ nmi}^{-2}$. These translate to 0 to 704 1000 mt, which are comparable to the MOCNESS-based biomass estimates of 588 and 839 1000 mt. Thus, while local estimates are commensurate, in this case, extrapolating the net-based samples to a much larger region is not advisable.

Regional estimates of krill density and abundance are available using the bongo-net samples from the NEFSC Ecomon surveys. *M. norvegica* are thought to easily avoid the Ecomon bongo net and density measurements of this species are thought to be underestimated. The extent of this underestimation is unknown, but Wiebe et al. (2013) showed that avoidance by *M. norvegica* to a larger MOCNESS net (1 m^2 vs. 0.61 m^2 for the bongo net) without a strobe light flashing is substantial (nearly an order of magnitude difference in catch and longer length classes between a strobe light flashing or not), and it is thought that this result can be extended to bongo nets—although this assertion has not been tested. When Euphausiacea densities were scaled proportional to the species composition of the net samples, the Ecomon-based biomass estimates were nearly an order of magnitude lower than the acoustic and the MOCNESS-based estimates. This level of mismatch is commensurate with the above evidence and comparisons of biomass estimates from the acoustic data and bongo catches suggest initial estimates of bongo catchability of *M. norvegica* of 0.09–0.36; but note that this does not account for size selectivity (the other component of catchability).

For the DWBA-assisted Z-score method, the DWBA-predicted reference means and standard deviations were implemented in the decibel (aka dB) domain rather than in the linear domain. The two main reasons for this are that the classification is done on the $S_v(f)$ data (i.e., dB domain) and the Z-score relies on normally distributed $S_v(f)$ and $\Delta S_v(f)$ data. This may seem counterintuitive because the scattering physics is in the linear domain where volume backscatter is calculated in linear form then converted to decibels. However, the distribution of s_v ($\text{m}^2 \text{ m}^{-3}$) is often skewed toward small values (Stanton 1985; Stanton and Clay 1986), so that the $\log_{10}$ transformation of s_v provides approximately normally distributed $S_v(f)$ values. In addition, the classification is done in the dB domain, so the reference values should be calculated in $\log_{10}$ domain.

Our initial intention was to use annual length information to estimate biomass on an annual basis (sensu, Reiss et al. 2008), but the paucity of length data for *M. norvegica* (and other Euphausiacea) in the Georges Bank region, and Gulf of Maine in general, forced us to combine length distributions from a variety of sources, but only from a few years. The combined length distribution was then incorporated in to a single weighted-mean length used for all survey years, and hence single length-weighted mean TS to convert acoustic energy to abundance and a single length-weighted mean weight to scale abundance to biomass that was applied to all the years. For a species that is considered to be an important, if not critical, component of the ecosystem (Cleary et al. 2012), the scarcity of data suggests highlights the overwhelming lack of resources for independent monitoring of krill. The importance of euphausiids in the ecosystem has been based on diet studies (e.g., Smith and Link 2010), which show many trophic levels (e.g., fish, birds, marine mammals) feed on euphausiids (Link et al. 2008). Without appropriate and adequate sampling, the role of *M. norvegica* in the Georges Bank and Gulf of Maine ecosystems can not be quantified. For example, the NEFSC ecosystem models place euphausiids in a micronekton “box” whose abundance and biomass is parameterized by bongo samples and whose consumption is parameterized by larval cod (Link et al. 2008). A need clearly exists for improved monitoring of this important species.

Conclusions

Ultimately choosing a classification method will depend on what information and data is available. All methods provided estimates that were within a factor of 3 of each other, and each method generally estimated higher or lower biomass relative to the others. Each method requires a priori information, but the number of parameters and level of detail differs among methods. The DWBA-assisted Z-score method is attractive because it has theoretical and empirical bases for developing the classification criteria, which should increase confidence in the results of the classification. However, incorporating

model-based frequency responses does come with the cost of a validated acoustic backscatter model over the frequencies and for the species of interest. The MFSBI and DWBA-assisted Z-score methods gave the lowest (i.e., conservative) estimates. This may reflect a more accurate representation of krill biomass, but considerably more validation needs to be done before a conclusive statement can be made. In addition, these two methods had the highest proportional biomass-to-standard error ratios (i.e., signal to noise ratio). The empirically based Z-score method requires clean (i.e., single species), validated training data, which can be difficult to obtain in situ, but does have the advantage that inter-survey changes in the backscattering properties of the organisms are accounted for. The dB-differencing and MFI methods gave biomass estimates in the middle of the other methods using literature-derived criteria (e.g., acceptance criteria were used directly from Trenkel and Berger [2013]). Potentially universal criteria suggest a common scattering process for krill in the North Atlantic. Whether this is true for other species remains an open question.

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Conflict of Interest

None declared.

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